

ECON 0100 | Fall 2024

MiniExam Z | Version 1

This MiniExam will take 20 minutes with quick break to follow. This exam covers key concepts from Parts A through E of the course. Answer each question clearly and completely.

Academic Conduct Code | Name: TAYLOR **Student ID:** J

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$$10A = 2E$$
$$5A = 1E$$

	A	E
O	1E	5A
G	1E	1A

Wandmakers Dilemma

Ollivander and Gregorovitch can each produce two very different types of wands.

Q1 | Production and Trade

In one day, Ollivander can make 10 ash wands (A) or 2 elder wands (E), while Gregorovitch can make 4 ash wands (A) or 4 elder wands (E).

- a) What is Ollivander's opportunity cost of making one elder wand (include units)? 5A
- b) What is Gregorovitch's opportunity cost of making one ash wand (include units)? 1E
- c) Who has the comparative advantage in making ash wands (include units)? Ollivander
- d) Circle all exchange rates that could facilitate trade between the wandmakers:
- 1 elder wand for 2 ash wands
 - 1 elder wand for 3 ash wands
 - 1 elder wand for 4 ash wands
 - 1 elder wand for 6 ash wands

Q2 | Market and Externalities

The market for ash wands can be represented by:

$$D : P = 120 - Q_D \quad (1)$$

$$S : P = 20 + Q_S \quad (2)$$

a) What is the market equilibrium quantity? 50

b) What is the market equilibrium price? 70

However, the process of treating wandwood releases polluted water into the local river, causing environmental damage estimated at 20 Galleons per wand produced.

c) What is the socially efficient quantity? 40

d) What corrective policy would achieve this quantity? Tax

e) What should the size of this corrective policy be? 20

$$120 - Q = 20 + Q$$

$$Q = 50$$

$$P = 70$$

$$MSB = MSC$$

$$120 - Q = 20 + Q + 20$$

$$2Q = 80$$

$$Q = 40$$

Q3 | Common Resource Problem

Two wandmakers share access to a magical forest where they gather wandwood. Each must decide whether to harvest sustainably or over-harvest. The payoff matrix shows their daily profits in Galleons:

		Gregorovitch		(3)
		S	OH	
Ollivander	Sustainable (S)	6, 7	1, 9	
	Over-Harvest (OH)	7, 0	2, 1	

a) What is Ollivander's best response when Gregorovitch chooses Sustainable? OH

b) What is Ollivander's best response when Gregorovitch chooses Over-harvest? OH

c) What is/are the Nash Equilibrium/Equilibria? (OH, OH)

d) What is the socially efficient outcome? (S, S)

Q4 | Market Structure

Currently, only Ollivander sells elder wands in Britain. If Gregorovitch expands his business to Britain, creating competition, circle the expected change in each of the following:

Price: UP, DOWN, SAME

Quantity: UP, DOWN, SAME

Deadweight Loss: UP, DOWN, SAME

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MiniExam Z | Version 2

This MiniExam will take 20 minutes with quick break to follow. This exam covers key concepts from Parts A through E of the course. Answer each question clearly and completely.

Academic Conduct Code | Name: TAYCOT **Student ID:** 😊

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	A	E
O	$\frac{1}{4}E$	4A
G	1E	1A

Wandmakers Dilemma

Ollivander and Gregorovitch can each produce two very different types of wands.

Q1 | Production and Trade

In one day, Ollivander can make 8 ash wands (A) or 2 elder wands (E), while Gregorovitch can make 3 ash wands (A) or 3 elder wands (E).

- a) What is Ollivander's opportunity cost of making one elder wand (include units)? 4A
- b) What is Gregorovitch's opportunity cost of making one ash wand (include units)? 1E
- c) Who has the comparative advantage in making ash wands (include units)? Ollivander
- d) Circle all exchange rates that could facilitate trade between the wandmakers:

1 elder wand for 2 ash wands

1 elder wand for 3 ash wands

1 elder wand for 4 ash wands

1 elder wand for 5 ash wands

Next time use
a different
number

Q2 | Market and Externalities

The market for ash wands can be represented by:

$$D : P = 100 - Q_D$$

$$S : P = 20 + Q_S$$

$$\begin{aligned} 100 - Q &= 20 + Q \\ 80 &= 2Q \\ Q &= 40 \\ P &= 60 \end{aligned}$$

(1)

(2)

a) What is the market equilibrium quantity? 40

b) What is the market equilibrium price? 60

However, the process of treating wandwood releases polluted water into the local river, causing environmental damage estimated at 30 Galleons per wand produced.

c) What is the socially efficient quantity? 25

d) What corrective policy would achieve this quantity? TAX

e) What should the size of this corrective policy be? 30

This one was short. Maybe do but next time.

$$\begin{aligned} MSB &= MSB \\ 100 - Q &= 20 + Q + 30 \\ 50 &= 2Q \\ Q &= 25 \end{aligned}$$

Q3 | Common Resource Problem

Two wandmakers share access to a magical forest where they gather wandwood. Each must decide whether to harvest sustainably or over-harvest. The payoff matrix shows their daily profits in Galleons:

Maybe try to make this one more interesting.

		Gregorovitch		(3)
		S	OH	
Ollivander	Sustainable (S)	6, 7	1, 9	
	Over-Harvest (OH)	7, 0	2, 1	

a) What is Ollivander's best response when Gregorovitch chooses Sustainable? OH

b) What is Ollivander's best response when Gregorovitch chooses Over-Harvest? OH

c) What is/are the Nash Equilibrium/Equilibria? (OH, OH)

d) What is the socially efficient outcome? (S, S)

Q4 | Market Structure

Currently, only Ollivander sells elder wands in Britain. If Gregorovitch expands his business to Britain, creating competition, circle the expected change in each of the following:

Price: UP, DOWN, SAME

Quantity: UP, DOWN, SAME

Deadweight Loss: UP, DOWN, SAME